

CSCI 3052U: Machine Learning I — Group Project Proposal (Milestone 1)

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1. Title

Evaluating Classical Machine Learning Baselines on Cantonese Natural Language Understanding

2. Abstract

Cantonese remains relatively under-resourced in natural language processing compared with higher-resource languages. The CantoNLU benchmark was created to evaluate Cantonese natural language understanding across seven tasks and reports results for several pretrained language models. This project will investigate how well simpler machine-learning approaches perform on selected CantoNLU tasks. We plan to focus initially on language detection and sentiment analysis, with a third task, creating a small pilot LLM, considered only if time and resources permit. We will compare naive benchmarks such as majority-class and stratified-random prediction with classical models including Multinomial Naive Bayes and logistic regression using character-level TF-IDF features. We will additionally evaluate one embedding-based comparison model — k-nearest neighbours on fixed multilingual sentence embeddings — which requires inference only, no fine-tuning. Models will be evaluated under a consistent train/validation/test procedure, with macro-F1 as the proposed primary metric (justified by cross-task consistency and per-class visibility rather than class imbalance, since the sentiment set is deliberately balanced) and accuracy and per-class measures as secondary metrics. The goal is not necessarily to outperform pretrained language models, but to establish clear, reproducible reference baselines and determine how much performance is gained from more complex pretrained representations. The final project will include a reproducible pipeline, fair model comparison, error analysis, robustness or sensitivity analysis, and a discussion of limitations. If the core comparison completes on schedule, a small pilot evaluation of one generative LLM on one task is retained as a stretch goal aligned with the Lee Lab's stated research direction.

3. Research Background

Natural language understanding (NLU) evaluates whether models can correctly interpret linguistic information in text. The Lee Language Lab's CantoNLU project provides a benchmark containing seven Cantonese NLU tasks and reports results from existing and newly trained language models. The broader CantoNLU research project is interested in extending evaluation to additional large language models and human performance.

For CSCI 3052U, we propose a narrower and more manageable question: how well can classical machine-learning models perform on selected CantoNLU tasks compared with existing pretrained-model results? Simple baselines provide an important reference point. If a classical model performs close to a pretrained model, this may suggest that useful task-specific patterns can be captured without large-scale pretraining. If the gap is large, it helps quantify the value of pretrained representations. This makes the project useful regardless of which model family performs best.

The main gap we will investigate is the absence of a simple naive-and-classical reference layer in our CantoNLU comparison. Earlier Cantonese sentiment-classification work also motivates the use of character-level features, while modern multilingual models provide a natural higher-complexity reference point. Our project therefore contributes the first missing reference layer of the lab's benchmarking agenda — reproducible naive and classical baselines — while explicitly leaving human-performance measurement to the lab's parallel annotation effort. A small pilot LLM evaluation (one model, one task) is retained as a stretch goal toward the lab's first stated direction, keeping the project substantively aligned with the approved Lee Lab problem.

Research Questions

- RQ1. How close do naive and classical feature-based models come to published CantoNLU pretrained-model results on selected tasks?
- RQ2. Does a fixed multilingual embedding representation improve performance over character-level TF-IDF without requiring full model fine-tuning?
- RQ3. How does model performance change with training-set size, and what types of examples are most often misclassified?

4. Methodology

Data

We plan to use released CantoNLU task datasets, initially focusing on language detection and sentiment analysis. A third task may be included only if the first two are completed successfully. Exact dataset versions, licenses, class distributions,

units of analysis, and official train/validation/test splits will be verified during Milestone 2 rather than assumed at Milestone 1. Preliminary inspection of the public repository confirms that official sentiment splits ship with the benchmark (approximately 10,000 / 1,000 / 1,000 rows; three deliberately balanced classes: smile, ok, cry), while the language-detection dataset is derived from public sources (e.g., Cantonese Wikipedia) rather than shipped ready-made; we will request the authors’ exact split and otherwise construct a documented stratified split for instructor approval. Because the repository currently carries no LICENSE file, written confirmation of course use will be requested from the benchmark authors and recorded in the data card.

Preprocessing and Models

Text will initially be represented with character-level TF-IDF features, which avoids requiring a separate Cantonese word-segmentation pipeline. We will inspect the data for duplicates, missing or invalid labels, class imbalance, and possible leakage. Our naive benchmarks will include majority-class and stratified-random prediction. Our primary course-level models will be Multinomial Naive Bayes and L2-regularized logistic regression. We will also evaluate k-nearest neighbours using fixed multilingual sentence embeddings (e.g., LaBSE); this requires inference only and runs on a laptop, so it is a committed comparison model rather than a conditional one. More complex models will remain optional until dataset size, compute needs, and feasibility are confirmed.

Experimental Design and Evaluation

Where official CantoNLU splits are available, we will use them; otherwise, we will propose an appropriate stratified split for instructor approval. Training data will be used to fit models, validation data will be used for model selection and hyperparameter tuning, and the final test set will be reserved for final evaluation. Hyperparameter-search effort will be documented and kept reasonably comparable across models. Published CantoNLU results will be quoted only where our splits and metrics match the benchmark’s protocol; otherwise they will be reported as indicative context rather than direct comparisons.

The proposed primary metric is macro-F1, chosen for cross-task consistency and per-class visibility (the sentiment task is class-balanced, so accuracy is reported alongside it), with accuracy, per-class precision and recall, and confusion matrices as secondary evidence. We will report variability across repeated runs where results are stochastic and will include at least one sensitivity, robustness, or ablation analysis. A likely sensitivity experiment is a learning curve using progressively larger portions of the training data. Error analysis will examine representative misclassified examples and discuss limitations such as domain coverage, dialect coverage, and any dependence on translation or external linguistic tooling.

Success Criterion

The project will be considered successful if we produce a reproducible data and training pipeline, naive benchmark results, at least two substantive machine-learning model results, a fair comparison on at least two CantoNLU tasks, uncertainty or variability evidence where appropriate, and a clear error-and-limitations analysis. Success does not depend on outperforming pretrained models. The pilot LLM evaluation is a stretch goal only and does not affect this success criterion.

5. Significance

This project can provide useful reference points for understanding how much performance on Cantonese NLU tasks can be achieved using relatively simple and computationally inexpensive methods. If classical models are competitive on some tasks, they may offer practical low-compute alternatives; if they fall substantially behind, the size of that gap helps quantify the value of pretrained language representations. The work is especially relevant to low-resource-language NLP, where data and compute constraints can make simpler methods valuable. It also fits the goals of Machine Learning I by requiring data preparation, baseline construction, model training, validation, evaluation, reproducibility, error analysis, and evidence-based conclusions.

6. SMART Timeline

Milestone	Specific, measurable target	Due
M1	Finalize proposal, team contract, roles, repository, and Lee Lab topic declaration.	Sept. 18
M2	Verify datasets/licenses; complete initial EDA; confirm tasks, splits, primary metric, risks, and compute plan. Confirm language-detection data source and	Oct. 2

	request written license/permission from the benchmark authors.	
M3	Build end-to-end pipeline; implement naive benchmark and first classical baselines; record seeds/environment and baseline table.	Oct. 19
M4	Present preliminary results and theory; identify blockers/deviations; revise timeline and respond to feedback. If on schedule: pilot LLM run (one model, one task; stretch).	Oct. 30
M5	Complete fair model comparison, tuning, variability analysis, error analysis, and one robustness/sensitivity/ablation study.	Nov. 20
M6	Complete 6-8 page report, reproducible repository, main-result regeneration, presentation/demo, contribution and AI-use statements.	Dec. 4
M7	Complete individual reflection, contribution evidence, and oral/code walkthrough.	Dec. 7

7. References

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